

Ondansetron: Drug information - UpToDate

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Ondansetron: Drug information

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For additional information see "Ondansetron: Patient drug information" and "Ondansetron: Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

Brand Names: US

- Zuplenz [DSC]

Brand Names: Canada

- ACCEL-Ondansetron;
- ACCEL-Ondansetron ODT;
- APO-Ondansetron;
- ATHENA-Ondansetron ODT;
- AURO-Ondansetron;
- Auro-Ondansetron ODT;
- CCP-Ondansetron [DSC];
- EUGIA-Ondansetron;
- JAMP Ondansetron;
- JAMP-Ondansetron;
- JAMP-Ondansetron ODF;
- Mar-Ondansetron;
- MAR-Ondansetron ODT;
- MINT-Ondansetron;
- MINT-Ondansetron ODT;
- MYLAN-Ondansetron;
- NAT-Ondansetron;
- Ondansetron ODT;
- Ondissolve ODF;
- PMS-Ondansetron;
- PMS-Ondansetron ODT;
- SANDOZ Ondansetron;
- TEVA Ondansetron;
- Zofran ODT

Pharmacologic Category

- Antiemetic;
- Selective 5-HT₃ Receptor Antagonist

Dosing: Adult

Dosage guidance:

Safety: Single IV doses >16 mg are no longer recommended due to the potential for QT prolongation (Ref). Avoid use in patients with congenital long-QT syndrome.

Carcinoid syndrome-associated diarrhea, severe, refractory

Carcinoid syndrome- associated diarrhea, severe, refractory (alternative agent) (off-label use): Based on limited data (case reports):

Oral: 8 mg 3 times daily (Ref) **or** 8 mg twice daily for 3 days, followed by a maintenance dose of 4 to 8 mg/day for 4 to 12 weeks (Ref).

IV: 4 to 8 mg every 8 hours (Ref).

Chemotherapy-induced nausea and vomiting, prevention

Chemotherapy-induced nausea and vomiting, prevention:

Single-day IV chemotherapy regimens:

Highly emetogenic chemotherapy (>90% risk of emesis [eg, cisplatin, breast cancer regimens that include an anthracycline combined with cyclophosphamide]):

Day of chemotherapy: Administer prior to chemotherapy **and** in combination with a neurokinin 1 (NK₁) receptor antagonist, dexamethasone, and olanzapine (Ref).

IV: 8 mg or 0.15 mg/kg as a single dose (Ref). Maximum: 16 mg/dose (Ref).

Oral:

Tablet formulations and oral solution: 8 mg twice daily for 2 doses with the first dose administered prior to chemotherapy administration (Ref) **or** 24 mg as a single dose (Ref).

Oral soluble film (Zuplenz: discontinued in the United States for >1 year): 24 mg (three 8 mg doses given together) as a single dose (Ref).

Post-chemotherapy days: 5-HT₃ receptor antagonist use is **not** necessary (other components of the antiemetic regimen are administered) (Ref).

Moderately emetogenic chemotherapy (30% to 90% risk of emesis): Carboplatin-based regimens:

Day of chemotherapy: Administer prior to chemotherapy **and** in combination with an NK₁ receptor antagonist and dexamethasone (Ref).

IV: 8 mg or 0.15 mg/kg as a single dose (Ref). Maximum: 16 mg/dose (Ref).

Oral: 8 mg twice daily for 2 doses with the first dose administered prior to chemotherapy administration, and the second dose 8 hours after the first dose (Ref).

Post-chemotherapy days: 5-HT₃ receptor antagonist use is not necessary (other components of the antiemetic regimen may be administered) (Ref).

Moderately emetogenic chemotherapy (30% to 90% risk of emesis): Non-carboplatin-based regimens:

Note: ASCO guidelines and MASCC/ESMO guidelines do not state a preference for which 5-HT₃ receptor antagonist should be used in this setting; however, palonosetron may be preferred (Ref).

Day of chemotherapy: Administer prior to chemotherapy **and** in combination with dexamethasone (Ref).

IV: 8 mg or 0.15 mg/kg as a single dose (Ref). Maximum: 16 mg/dose (Ref).

Oral: 8 mg twice daily for 2 doses with the first dose administered prior to chemotherapy administration, and the second dose 8 hours after the first dose (Ref).

Post-chemotherapy days: 5-HT₃ receptor antagonist use is **not** necessary (other components of the antiemetic regimen may be administered) (Ref); however, if a first-generation 5-HT₃ receptor antagonist (eg, ondansetron, granisetron) was used on day 1 of chemotherapy rather than palonosetron, the first-generation 5-HT₃ receptor antagonist may be continued for post-chemotherapy emetic prophylaxis on days 2 and 3 (Ref).

Low emetogenic risk (10% to 30% risk of emesis):

Note: Single-agent ondansetron is an option for prophylaxis (Ref).

Day of chemotherapy:

IV: 8 mg as a single dose prior to chemotherapy (Ref).

Oral (off-label): 8 mg as a single dose prior to chemotherapy (Ref).

Post-chemotherapy days: Prophylaxis is not necessary on subsequent days (Ref).

Minimal emetogenic risk (<10% risk of emesis): Routine antiemetic prophylaxis is not generally necessary (Ref).

High-dose chemotherapy with stem or bone marrow transplant:

Day of chemotherapy: Administer prior to chemotherapy **and** in combination with a neurokinin 1 (NK₁) receptor antagonist, dexamethasone, with or without olanzapine (Ref).

IV: 8 mg or 0.15 mg/kg as a single dose (Ref). Maximum: 16 mg/dose (Ref).

Oral: 24 mg as a single dose (Ref).

Oral chemotherapy agents:

High/moderate emetogenic risk oral agent (≥30% risk of emesis): Oral: 8 to 16 mg/day administered before chemotherapy and continued daily (Ref).

Low/minimal emetogenic risk oral agent (<30% risk of emesis): Oral: 8 to 16 mg/day on an as-needed basis only (Ref).

Gastroparesis, symptomatic treatment of nausea and vomiting

Gastroparesis, symptomatic treatment of nausea and vomiting (alternative agent) (off-label use):

Note: For patients with persistent symptoms refractory to prokinetic therapy. No data available; recommendations for use and dose are based on expert opinion.

Oral: 4 to 8 mg 3 times daily (Ref).

Nausea and/or vomiting, acute, severe

Nausea and/or vomiting, acute, severe (off-label use):

Note: Use has primarily been evaluated in patients with undifferentiated nausea/vomiting presenting to the emergency department; however, may also use for nausea/vomiting due to viral gastroenteritis, acute mountain sickness, cyclical vomiting syndrome, palliative care, and a variety of other medical conditions associated with severe, self-limiting acute nausea/vomiting (Ref).

Oral, IV, IM: 4 to 8 mg as a single dose (Ref); may repeat 4 to 8 mg every 4 to 8 hours as needed (Ref). **Note:** For parenteral therapy, IV administration is preferred over IM when possible (Ref).

Nausea and vomiting, pregnancy associated, severe or persistent

Nausea and vomiting, pregnancy associated, severe or persistent (off-label use):

Note: May be considered as an adjunctive treatment of nausea and vomiting when symptoms are severe or persist following initial pharmacologic therapy (Ref).

Patients without hypovolemia: Oral, IV (bolus): 4 mg every 8 hours, as needed, added to current treatment regimen (Ref). If necessary, some experts increase to 8 mg every 8 to 12 hours, as needed (Ref).

Patients with hypovolemia:

Note: For patients with persistent symptoms despite intravenous fluid replacement:

IV: 8 mg administered over 15 minutes every 12 hours, added to current treatment regimen (Ref). Some experts use 4 to 8 mg administered as an IV bolus every 8 hours until stabilization (Ref).

Postoperative nausea and vomiting, prevention

Postoperative nausea and vomiting, prevention:

Moderate- to high-risk patients:

Note: In patients at moderate risk, may combine ondansetron with other prophylactic interventions (eg, another antiemetic agent from a different pharmacologic class, modification of anesthetic technique, acupuncture); in patients at high risk, combine 3 or more interventions (Ref).

Usual dose: IV: 4 mg as a single dose at the end of surgery (Ref).

Alternative strategy: Oral (oral disintegrating tablet or oral soluble film [Zuplenz: discontinued in the United States for >1 year]): 8 mg as a single dose given 30 to 60 minutes prior to surgery (Ref).

Low-risk patients: Although prophylaxis is not always indicated in low-risk patients, consensus guidelines acknowledge that some experts may administer an antiemetic in these patients; however, clinicians are also advised that this strategy comes with the potentially unnecessary risk of rare adverse effects (Ref). If ondansetron is given, the dosing is the same as for moderate- to high-risk patients.

Post-discharge management in high-risk patients: Limited data available; dosage regimen studied in a single clinical trial:

Oral (oral disintegrating tablet or oral soluble film [Zuplenz: discontinued in the United States for >1 year]): 8 mg to be taken on discharge and in the morning of postoperative days 1 and 2 (Ref).

Postoperative nausea and vomiting, treatment or rescue therapy

Postoperative nausea and vomiting, treatment or rescue therapy (off-label use):

Note: Rescue therapy should always include an antiemetic from a different class than the one used for prophylaxis, unless a potentially inadequate dose was initially administered or the effect of the first drug has worn off (>6 hours since initial dose for most 5-HT₃ receptor antagonists) (Ref). However, some experts do not recommend repeat administration of a 5-HT₃ antagonist unless no alternatives are available for rescue (Ref).

IV: 4 mg as a single dose when a prophylactic agent was not utilized (treatment) or following failure of an agent utilized as prophylaxis (rescue therapy) (Ref).

Oral (oral disintegrating tablet or oral soluble film [Zuplenz: discontinued in the United States for >1 year]): 4 or 8 mg as a single dose when a prophylactic agent was not utilized (treatment) or following failure of an agent utilized as prophylaxis (rescue therapy) (Ref).

Radiation therapy-associated nausea and vomiting, prevention

Radiation therapy-associated nausea and vomiting, prevention:

High-emetogenic risk radiation therapy (total body irradiation):

Radiation day(s):

IV (off-label): 8 mg or 0.15 mg/kg (maximum: 16 mg/dose)(Ref) once daily or twice daily prior to each fraction of radiation; administer in combination with dexamethasone (Ref).

Oral: 8 mg once daily or twice daily administered 1 to 2 hours prior to each fraction of radiation; administer in combination with dexamethasone (Ref) **or**, in one clinical trial of 4 days of hyperfractionated total body irradiation, 8 mg (without dexamethasone) was administered 1.5 hours prior to every fraction of radiation (3 times daily for the first 3 days and twice daily on day 4) (Ref).

Post-radiation days:

IV (off-label), Oral: The appropriate duration of therapy following radiotherapy days is not well defined; ASCO guidelines recommend continuing ondansetron once daily or twice daily on the day after each day of radiation (Ref).

Moderate-emetogenic risk radiation therapy (upper abdomen, craniospinal irradiation) (off-label use):

Radiation day(s):

IV (off-label): 8 mg **or** 0.15 mg/kg (maximum: 16 mg/dose [manufacturer's labeling]) once daily or twice daily prior to each fraction of radiation; may administer with or without dexamethasone before the first 5 fractions (Ref).

Oral: 8 mg once daily or twice daily administered 1 to 2 hours prior to each fraction of radiation; may administer with or without dexamethasone before the first 5 fractions (Ref) **or**, in clinical trials involving upper abdomen radiation (high-dose single exposure or multiple-day fractionated course), 8 mg 3 times daily (without dexamethasone) has been given; doses were administered 1 to 2 hours prior to radiation therapy (Ref).

Low- (brain, head and neck, thorax, pelvis) to minimal- (extremities, breast) emetogenic risk radiation therapy: Routine prophylaxis **not** recommended; however, may use as rescue therapy using the following dosing with consideration of using prophylactically for the remainder of radiation therapy (Ref).

IV: 8 mg or 0.15 mg/kg (Ref) (maximum: 16 mg/dose (Ref)).

Oral: 8 mg (Ref).

Vertigo-associated nausea and vomiting

Vertigo-associated nausea and vomiting (alternative agent) (off-label use):

IV (preferred), IM: 4 to 8 mg once for acute symptoms (Ref).

Oral: 4 mg every 8 to 12 hours as needed (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

The renal dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.

Altered kidney function: IV, Oral: No dose adjustments likely to be necessary, as clearance by the kidney accounts for only 5% of total clearance (Roila 1995; manufacturer's labeling). Unlikely to be significantly dialyzed due to relatively high volume of distribution and plasma protein binding (Ref).

Dosing: Liver Impairment: Adult

The liver dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Matt Harris, PharmD, MHS, BCPS, FAST, FCCP; Jeong Park, PharmD, MS, BCTXP, FCCP, FAST; Arun Jesudian, MD; Sasan Sakiani, MD.

Note: Small pharmacokinetic studies comparing healthy controls to patients with cirrhosis demonstrate significant changes in elimination pharmacokinetics. In mild to moderate liver cirrhosis, the

elimination half-life doubles and clearance is reduced by 50%. In severe cirrhosis, the half-life increases >300% and clearance is reduced by ~80% (Ref). Accumulation of ondansetron can yield constipation, QT_c prolongation, and headache (Ref).

Liver impairment prior to treatment initiation:

Initial or dose adjustment in patients with preexisting liver cirrhosis:

Child-Turcotte-Pugh class A and B: IM, IV, Oral: No dosage adjustment necessary (Ref).

Child-Turcotte-Pugh class C: IM, IV, Oral: Administer 4 to 8 mg every 12 to 24 hours; maximum dose: 8 mg per 24 hours (Ref).

Dosing: Adjustment for Toxicity: Adult

Hypersensitivity: Discontinue ondansetron; manage as clinically indicated.

Serotonin syndrome: Discontinue ondansetron and initiate supportive management.

Dosing: Older Adult

Oral: No dosing adjustment required; refer to adult dosing.

IV: Single IV doses >16 mg are no longer recommended due to the potential for QT prolongation (Ref). In patients ≥75 years, Canadian recommendations place additional restrictions to limit initial IV doses to ≤8 mg due to this risk (Ref).

Dosing: Pediatric

(For additional information see "Ondansetron: Pediatric drug information")

Chemotherapy-induced nausea and vomiting, prevention

Chemotherapy-induced nausea and vomiting, prevention:

Note: Use in combination with or without dexamethasone and aprepitant or fosaprepitant depending upon patient age, chemotherapy emetogenic potential, and drug-interaction profile; dosing regimens variable, refer to specific protocols or institutional guidelines (Ref).

Multiple-daily dose regimen (high, moderate, low emetic potential):

Weight-directed dosing:

Infants, Children, and Adolescents: IV, Oral: 0.15 mg/kg/dose (5 mg/m²/dose); maximum single dose: 16 mg/dose; administer first dose before the start of chemotherapy and then every 4 to 12 hours, not to exceed 3 doses; maximum daily dose: 0.45 mg/kg/day or 32 mg/day; whichever is less; higher emetogenic potential chemotherapy will require more frequent administration (Ref).

Fixed dosing:

Children 4 to 11 years: Oral: 4 mg beginning 30 minutes before chemotherapy; repeat 4 and 8 hours after initial dose, then 4 mg every 8 hours for 1 to 2 days after chemotherapy completed.

Children ≥12 years and Adolescents: Oral: 8 mg beginning 30 minutes before chemotherapy; repeat dose 8 hours after initial dose, then 8 mg every 12 hours for 1 to 2 days after chemotherapy completed.

Single-daily dose regimen (low, moderate, or highly emetogenic potential): Limited data available:

Infants, Children, and Adolescents: IV: 0.3 to 0.45 mg/kg/dose once daily; maximum dose: 16 mg/dose (Ref).

Cyclic vomiting syndrome, supportive/rescue therapy

Cyclic vomiting syndrome, supportive/rescue therapy: Limited data available:

Children and Adolescents: IV: 0.3 to 0.4 mg/kg/dose every 4 to 6 hours as needed; maximum dose: 8 mg/dose (Ref). Maximum daily dose: 32 mg/day (Ref).

Gastroenteritis, acute; treatment

Gastroenteritis, acute; treatment: Limited data available (Ref):

IV: Infants and Children: IV: 0.15 or 0.3 mg/kg/dose once; maximum dose: 16 mg/dose (Ref).

Oral: **Note:** May repeat dose if patient vomits within 15 minutes (Ref).

Infants ≥ 3 months, Children, and Adolescents:

<15 kg: Oral: 0.2 mg/kg/dose once (Ref).

15 to 30 kg: Oral: 4 mg once (Ref).

>30 kg: Oral: 8 mg once (Ref).

Postoperative nausea and vomiting, prevention

Postoperative nausea and vomiting, prevention: Note: Administer immediately before or following induction of anesthesia, or postoperatively if the patient is symptomatic. Repeat doses administered in response to inadequate control of nausea/vomiting from preoperative doses are generally ineffective (Ref).

Infants and Children: IV: 0.05 to 0.1 mg/kg/dose as a single dose; maximum dose: 4 mg/dose (Ref).

Adolescents: IM, IV: 4 mg/dose as a single dose (Ref).

Radiation-induced nausea and vomiting, prevention

Radiation-induced nausea and vomiting, prevention: Limited data available:

Weight-directed dosing: Infants ≥ 5 months, Children, and Adolescents: Oral: 0.2 mg/kg/dose (maximum dose: 8 mg/dose) administered every 8 hours throughout total body irradiation (TBI) prior to hematopoietic stem cell transplant (HSCT) (n=68; mean age: 6.7 years; range: 5 months to 20 years); doses were generally rounded to 4 mg/dose in children 4 to 11 years and 8 mg/dose in children ≥ 12 years and adolescents (Ref).

Alternate weight-based dosing: Children and Adolescents: Oral: 0.15 mg/kg/dose administered 3 to 4 times daily throughout TBI (n=33; mean age: 9 years; range: 13 months to 16 years) (Ref).

Fixed dose: **Note:** Derived from rounding weight-based (0.2 mg/kg/dose) doses (Ref).

Children 4 to 11 years: Oral: 4 mg every 8 hours throughout TBI prior to HSCT.

Children ≥ 12 years and Adolescents: Oral: 8 mg every 8 hours throughout TBI prior to HSCT.

Alternate fixed-dosing: Children ≥ 9 years and Adolescents: Oral: 8 mg every 12 hours on days of TBI prior to bone marrow transplantation (age range: 9 to 67 years; median age range: 39 to 49 years).
Note: Administered in combination with dexamethasone (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

Infants, Children, and Adolescents: IV, Oral: No dose adjustments likely to be necessary, as clearance by the kidney accounts for only 5% of total clearance (Ref).

Dosing: Liver Impairment: Pediatric

Infants, Children, and Adolescents: There are no pediatric-specific recommendations; based on experience in adult patients, no adjustment may be necessary for mild to moderate hepatic impairment; for severe impairment, dosing adjustment suggested.

Adverse Reactions (Significant): Considerations

Constipation

Onaansetron may cause **constipation** (Ref). **Intestinal obstruction** has also been reported (Ref).

Mechanism: Likely due to blockade of 5-HT₃ gut receptors, which results in decreased motility (Ref). May also occur due to slowed colonic transit time (Ref) and inhibition of postprandial increase in tone (Ref).

Onset: Varied; one reported case of constipation occurred the same day as oral ondansetron administration; intestinal obstruction resulted approximately 2 weeks after continuous twice daily dosing (Ref).

Risk factors:

- Preexisting liver disease (Ref)

Headache

Headache is the most reported adverse reaction with ondansetron (Ref). Additional doses and movement (with or without postural changes) have been associated with worsening (Ref). Severe headache has led to discontinuation in reported cases (Ref).

Onset: Rapid; severe headache reported within minutes of administration, often lasting minutes to hours. There have been multiple case reports of children experiencing throbbing headaches for several days following ondansetron and chemotherapy (Ref). One patient reported onset of severe headache occurring within 2 hours of the first dose and continued to occur every 12 hours following repeated administration (Ref).

Risk factors:

- Higher doses (potential risk factor) (Ref)
- Concurrent use of propofol and/or fentanyl (Ref)
- Personal or family history of migraine (Ref)

Hypersensitivity (immediate)

Immediate **hypersensitivity reactions** may occur with ondansetron, including **urticaria, angioedema, hypotension, bronchospasm, dyspnea, and anaphylaxis** (Ref). Some patients may only present with hypotension, without any accompanying symptoms (Ref). Patients may have more severe reactions on subsequent exposure (Ref).

Mechanism: Non-dose-related; immunologic; likely IgE-mediated (Ref); may be the result of direct mast cell stimulation (in some patients without previous exposure) (Ref).

Onset: Rapid; immediate hypersensitivity reactions generally occur within 1 hour of administration, but may occur up to 6 hours after exposure (Ref). Reactions usually occur after the first dose during the second or third course of chemotherapy (Ref).

Risk factors:

- Adults with cancer (Ref); however, a few isolated cases have been reported in children (Ref)
- IV route of administration; however, anaphylaxis following sublingual administration has been reported (Ref)
- Cross-reaction between serotonin 5-HT₃ antagonists has been described in limited case reports (Ref)

QT prolongation

Increases in ECG intervals (eg, PR, QRS duration, JT); **prolonged QT interval on ECG**; and **bradycardia** have been observed with ondansetron (Ref). Cases of ventricular arrhythmias and **torsades de pointes** have also been reported. Rare cases of fatalities have occurred even at low doses (Ref).

Mechanism: QT prolongation may occur due to HERG K⁺ channel-blockade (Ref). Suppression of autonomic reflexes may contribute to bradycardia, hypotension, and tachyarrhythmias (Ref).

Onset: Rapid; usually occurs 1 to 2 hours after administration (Ref); however, QT prolongation peaks have been observed within ~5 to 15 minutes following administration (Ref). QTc intervals >500

ms have been recorded within 15 minutes after administration (Ref). May persist >2 hours from administered dose (Ref).

Risk factors:

- IV route of administration (Ref)
- Single doses >16 mg IV (Ref); however, risk should be considered even with low IV doses (Ref)
- Concomitant medications that prolong the QT interval (Ref)
- Females (Ref)
- Hypothermia (Ref)
- Concomitant volatile anesthetics or cumulative high-dose anthracycline therapy (Ref)
- Underlying heart disease including heart failure or acute coronary syndromes (Ref)
- Electrolyte abnormalities (eg, hypokalemia, hypomagnesemia) (Ref)
- History of QT prolongation, bradycardia, tachycardia, or cardiac rhythm disorders (especially ventricular arrhythmia) (Ref)
- Patients receiving ondansetron following anesthesia, while in the intensive care unit, or during hospital admission (Ref)

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified. Incidence reported in adult patients unless otherwise specified.

>10%:

Gastrointestinal: Constipation (9% to 11%) (table 1)

Table 1: Ondansetron: Adverse Reaction

Drug (Ondansetron)	Placebo	Dose	Dosage Form	Indication
9% 11%	0.4% N/A	8 mg twice daily N/A	Oral Injection	Chemotherapy-induced nausea and vomiting N/A

Nervous system: Fatigue (oral: ≤13%), headache (9% to 24%) (table 2)

Drug (Ondansetron)	Comparator (Metoclopramide)	Placebo	Dose	Dosage Form
24%	N/A	13%	8 mg twice daily	Oral
17%	7%	15%	0.15 mg/kg x 3	Injection
17%	N/A	14%	4 mg single dose	IV
9%	N/A	5%	16 mg single dose	Oral

1% to 10%:

Dermatologic: Pruritus (2% to 5%), skin rash (1%)

Gastrointestinal: Diarrhea (oral: 6%; IV: children 1 to 24 months of age: 2%)

Genitourinary: Gynecologic disease (oral: 7%), urinary retention (oral: 5%)

Hepatic: Increased serum alanine aminotransferase (>2 times ULN: 1% to 5%; transient), increased serum aspartate aminotransferase (>2 times ULN: 1% to 5%; transient)

Hypersensitivity: Anaphylaxis (<2%) (Ref)

Local: Injection site reaction (4%; includes burning sensation at injection site, erythema at injection site, injection site pain)

Nervous system: Agitation (oral: ≤6%), anxiety (oral: ≤6%), dizziness (7%), drowsiness (IV: ≤8%), paresthesia (IV: 2%), sedated state (IV: ≤8%), sensation of cold (IV: 2%)

Respiratory: Bronchospasm (<2%), hypoxia (oral: 9%)

Miscellaneous: Fever (2% to 8%)

<1%:

Cardiovascular: Hypotension

Nervous system: Extrapyrarnidal reaction (Ref)

Frequency not defined:

Cardiovascular: Angina pectoris, peripheral vascular disease, tachycardia

Endocrine & metabolic: Hypokalemia

Nervous system: Tonic clonic epilepsy

Postmarketing:

Cardiovascular: Atrial fibrillation (Ref), bradycardia (Ref), depression of ST segment on ECG, flushing, ischemic heart disease (most commonly due to coronary artery spasm and may occur with oral or IV [predominantly IV]; occurred immediately after IV administration and resolved with treatment), palpitations, prolonged QT interval on ECG (Ref), second degree atrioventricular block, supraventricular tachycardia, syncope, torsades de pointes (Ref), ventricular premature contractions, ventricular tachycardia

Dermatologic: Stevens-Johnson syndrome, toxic epidermal necrolysis (Ref), urticaria (Ref)

Gastrointestinal: Hiccups, intestinal obstruction (Ref)

Hematologic & oncologic: Positive lymphocyte transformation test

Hepatic: Hepatic failure

Hypersensitivity: Angioedema, fixed drug eruption (Ref), hypersensitivity reaction (Ref), nonimmune anaphylaxis

Nervous system: Dystonic reaction (Ref), serotonin syndrome (Ref)

Neuromuscular & skeletal: Laryngospasm

Ophthalmic: Accommodation disturbance, oculogyric crisis (Ref), transient blindness (lasted ≤48 hours) (Ref), transient blurred vision (following infusion)

Respiratory: Dyspnea, laryngeal edema, stridor

Contraindications

Hypersensitivity to ondansetron or any component of the formulation; concomitant use with apomorphine

Warnings/Precautions

Concerns related to adverse effects:

- Serotonin syndrome: Serotonin syndrome (SS) has been reported with 5-HT₃ receptor antagonists, predominantly when used in combination with other serotonergic agents (eg, selective serotonin reuptake inhibitors, serotonin and norepinephrine reuptake inhibitors, monoamine oxidase inhibitors, mirtazapine, fentanyl, lithium, tramadol, and/or methylene blue). Some of the cases have been fatal. The majority of serotonin syndrome reports due to 5-HT₃ receptor antagonist have occurred in a postanesthesia setting or in an infusion center. SS has also been reported following overdose

of ondansetron. Signs/symptoms of SS include mental status changes (eg, agitation, hallucinations, delirium, coma); autonomic instability (eg, tachycardia, labile BP, diaphoresis, dizziness, flushing, hyperthermia); neuromuscular changes (eg, tremor, rigidity, myoclonus, hyperreflexia, incoordination); GI symptoms (eg, nausea, vomiting, diarrhea); and/or seizures.

Dosage form specific issues:

- Benzyl alcohol and derivatives: Some dosage forms may contain sodium benzoate/benzoic acid; benzoic acid (benzoate) is a metabolite of benzyl alcohol; large amounts of benzyl alcohol (≥ 99 mg/kg/day) have been associated with a potentially fatal toxicity (“gasping syndrome”) in neonates; the “gasping syndrome” consists of metabolic acidosis, respiratory distress, gasping respirations, CNS dysfunction (including convulsions, intracranial hemorrhage), hypotension, and cardiovascular collapse (AAP [“Inactive” 1997]; CDC 1982); some data suggests that benzoate displaces bilirubin from protein binding sites (Ahlfors 2001); avoid or use dosage forms containing benzyl alcohol derivative with caution in neonates. See manufacturer’s labeling.
- Phenylalanine: Orally disintegrating tablets contain phenylalanine.

Other warnings/precautions:

- Chemotherapy-associated emesis: Antiemetics are most effective when used prophylactically (MASCC/ESMO [Herrstedt 2024]). If emesis occurs despite optimal antiemetic prophylaxis, re-evaluate emetic risk, disease status, concurrent morbidities and current medications to assure antiemetic regimen is optimized (ASCO [Hesketh 2020]).

Product Availability

Zuplenz has been discontinued in the United States for >1 year.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Film, Oral:

Zuplenz: 4 mg (30 ea [DSC])

Solution, Injection, as hydrochloride [strength expressed as base]:

Generic: 4 mg/2 mL (2 mL); 40 mg/20 mL (20 mL)

Solution, Injection, as hydrochloride [strength expressed as base, preservative free]:

Generic: 4 mg/2 mL (2 mL)

Solution, Oral, as hydrochloride [strength expressed as base]:

Generic: 4 mg/5 mL (5 mL, 50 mL)

Solution Prefilled Syringe, Injection, as hydrochloride:

Generic: 4 mg/2 mL (2 mL)

Tablet, Oral, as hydrochloride [strength expressed as base]:

Generic: 4 mg, 8 mg, 24 mg

Tablet Disintegrating, Oral:

Generic: 4 mg, 8 mg, 16 mg

Generic Equivalent Available: US

May be product dependent

Pricing: US

Solution (Ondansetron HCl Injection)

4 mg/2 mL (per mL): \$0.15 - \$1.35

40 mg/20 mL (per mL): \$0.14 - \$1.25

Solution (Ondansetron HCl Oral)

4 mg/5 mL (per mL): \$4.79 - \$6.00

Solution Prefilled Syringe (Ondansetron HCl Injection)

4 mg/2 mL (per mL): \$1.11

Tablet, orally-disintegrating (Ondansetron Oral)

4 mg (per each): \$22.25 - \$23.11

8 mg (per each): \$36.66 - \$38.50

16 mg (per each): \$47.09

Tablets (Ondansetron HCl Oral)

4 mg (per each): \$23.63 - \$24.78

8 mg (per each): \$35.68 - \$41.53

24 mg (per each): \$106.51

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Film, Oral:

Ondissolve ODF: 4 mg (6 ea, 10 ea, 50 ea); 8 mg (6 ea, 10 ea, 50 ea) [contains levomenthol, polyethylene glycol (macrogol), polysorbate 80]

Generic: 4 mg (10 ea); 8 mg (10 ea)

Solution, Intravenous:

Generic: 2 mg/mL (2 mL, 4 mL, 5 mL, 20 mL)

Solution, Oral, as hydrochloride [strength expressed as base]:

Generic: 4 mg/5 mL (50 mL)

Tablet, Oral, as hydrochloride [strength expressed as base]:

Generic: 4 mg, 8 mg

Tablet Disintegrating, Oral:

Zofran ODT: 4 mg, 8 mg [contains aspartame, methylparaben sodium, propylparaben sodium]

Generic: 4 mg, 8 mg

Extemporaneous Preparations

Note: Commercial oral solution is available (0.8 mg/mL)

If commercial oral solution is unavailable, a 0.8 mg/mL syrup may be made with ondansetron tablets, Ora-Plus® (Paddock), and any of the following syrups: Cherry syrup USP, Syrpalta® (HUMCO), Ora-Sweet® (Paddock), or Ora-Sweet® Sugar-Free (Paddock). Crush ten 8 mg tablets in a mortar and reduce to a fine powder (flaking of the tablet coating occurs). Add 50 mL Ora-Plus® in 5 mL increments, mixing thoroughly; mix while adding the chosen syrup in incremental proportions to **almost** 100 mL; transfer to a calibrated bottle, rinse mortar with syrup, and add sufficient quantity of syrup to make 100 mL. Label "shake well" and "refrigerate". Stable for 42 days refrigerated (Ref).

Rectal suppositories: Calibrate a suppository mold for the base being used. Determine the displacement factor (DF) for ondansetron for the base being used (Fattibase® = 1.1; Polybase® = 0.6). Weigh the ondansetron tablet(s). Divide the tablet weight by the DF; this result is the weight of base displaced by the drug. Subtract the weight of base displaced from the calculated weight of base required for each suppository. Grind the ondansetron tablets in a mortar and reduce to a fine powder. Weigh out the appropriate weight of suppository base. Melt the base over a water bath (<55°C). Add the ondansetron powder to the suppository base and mix well. Pour the mixture into the suppository mold and cool. Stable for at least 30 days refrigerated (Ref).

Administration: Adult

Oral: Administer oral dosage forms 30 minutes prior to chemotherapy; 1 to 2 hours before radiation; 30 to 60 minutes prior to surgery or induction of anesthesia.

Orally disintegrating tablets: Do not remove from blister until needed. Peel backing off the blister, do not attempt to push tablet through the foil. Using dry hands, place tablet on tongue and allow to dissolve. Swallow with saliva (no need to swallow with liquids).

Oral soluble film: Do not remove from pouch until immediately before use. Using dry hands, place film on top of tongue and allow to dissolve (4 to 20 seconds). Swallow with or without liquid. If using more than one film, each film should be allowed to dissolve completely before administering the next film.

Enteral feeding tube:

The following recommendations are based upon the best available evidence and clinical expertise. Senior editorial team: Joseph I. Boullata, PharmD, RPh, CNS-S, FASPEN, FACN; Peggi A. Guenter PhD, RN, FASPEN; Kathleen Gura, PharmD, BCNSP, FASHP, FASPEN, FPPA, FMSHP; Mark G. Klang MS, RPh, BCNSP, PhD, FASPEN; Linda Lord, NP, ACNP-BC, CNSC, FASPEN.

Oral film:

Enteral feeding tube administration utilizing ondansetron oral films is not recommended (Ref).

Oral solution (commercially available):

Gastric (eg, NG, G-tube) or post-pyloric (eg, J-tube) tubes: Dilute dose with at least an equivalent volume of purified water prior to administering to reduce osmolality and viscosity; some experts recommend diluting in a volume of purified water that is 3 times the ondansetron solution volume (eg, 10 mL ondansetron solution diluted in up to 30 mL purified water). Draw up diluted solution into enteral dosing syringe and administer via feeding tube (Ref).

Dosage form information: One undiluted formulation has been reported to have an osmolality of ~2,900 mOsm/kg (Ref); oral solutions with an osmolality >600 mOsm/kg may increase the probability of adverse GI effects (eg, diarrhea, cramping, abdominal distention, slowed gastric emptying), particularly if administered post pylorically, inadequately diluted, and/or used in at-risk patients (eg, neonates and infants, patients with short bowel syndrome) (Ref).

General guidance: Hold enteral nutrition (EN) during ondansetron administration (Ref). Flush feeding tube with an appropriate volume of purified water (eg, 15 mL) before administration (Ref). Following administration, rinse container used for preparation with purified water; draw up rinse and administer contents to ensure delivery of entire dose (Ref). Flush feeding tube with an appropriate volume of purified water (eg, 15 mL) and restart EN (Ref).

Oral tablet:

Gastric (eg, NG, G-tube) or post-pyloric (eg, J-tube) tubes: Crush tablet(s) into a fine powder and disperse in at least 10 mL purified water immediately prior to administration; draw up mixture into enteral dosing syringe and administer via feeding tube (Ref).

Dosage form information: Some formulations may be film coated; administration of film-coated ondansetron tablets via feeding tube may increase the risk of clogging the tube; if used, ensure tablets are dispersed sufficiently with an adequate amount of purified water prior to administration (Ref).

General guidance: Hold EN during ondansetron administration (Ref). Flush feeding tube with an appropriate volume of purified water (eg, 15 mL) before administration (Ref). Following administration, rinse container used for preparation with purified water; draw up rinse and administer contents to ensure delivery of entire dose (Ref). Flush feeding tube with an appropriate volume of purified water (eg, 15 mL) and restart EN (Ref).

Oral tablet, disintegrating:

Gastric (eg, NG, G-tube) or post-pyloric (eg, J-tube) tubes: Remove plunger from an empty enteral dosing syringe and place orally disintegrating ondansetron tablet(s) into syringe barrel. Replace the syringe plunger and draw up an appropriate volume (eg, 10 to 30 mL) of purified water. Cap the end of the syringe and gently agitate to disperse tablet prior to administration via feeding tube (Ref). **Note:** Some experts suggest that post-pyloric administration is not preferred; consider an alternative dosage form if symptoms do not improve following post-pyloric administration (Ref).

General guidance: Hold EN during ondansetron administration (Ref). Flush feeding tube with an appropriate volume of purified water (eg, 15 mL) before administration (Ref). Following administration, rinse enteral dosing syringe used for preparation with purified water and administer contents to ensure delivery of entire dose (Ref). Flush feeding tube with an appropriate volume of purified water (eg, 15 mL) and restart EN (Ref).

Note: Recommendations may not account for differences in inactive ingredients, osmolality, or other formulation properties that may vary among products from different manufacturers.

IM: Administer undiluted.

IV:

IVPB: Infuse diluted solution over 15 minutes.

Chemotherapy-induced nausea and vomiting: Give first dose 30 minutes prior to beginning chemotherapy.

IV push: Prevention of postoperative nausea and vomiting: Single doses may be administered IV injection as undiluted solution over at least 30 seconds but preferably over 2 to 5 minutes.

Preparation for Administration: Adult

Vial or prefilled single use syringe:

Prevention of chemotherapy-induced nausea and vomiting: Dilution is required prior to IV infusion. Dilute in 50 mL D5W or NS. Use diluted solutions within 24 hours of preparation.

Prevention of postoperative nausea and vomiting: No dilution is required.

Administration: Pediatric

The following feeding tube recommendations are based upon the best available evidence and clinical expertise. Senior editor panel: Joseph I. Boullata, PharmD, RPh, CNS-S, FASPEN, FACN; Peggy A. Guenter, PhD, RN, FASPEN; Kathleen Gura, PharmD, BCNSP, FASHP, FASPEN, FPPA, FMSHP; Mark G. Klang, MS, RPh, BCNSP, PhD, FASPEN; Linda Lord, NP, ACNP-BC, CNSC, FASPEN.

Note: Recommendations may not account for differences in inactive ingredients, osmolality, or other formulation properties that may vary among products from different manufacturers.

Oral: May administer without regard to meals. For chemotherapy-induced nausea and vomiting, administer 30 minutes prior to chemotherapy.

Orally disintegrating tablet (Zofran ODT): Do not remove from blister until needed. Peel backing off the blister; do not push tablet through foil backing. Using dry hands, place tablet on tongue and allow to dissolve; swallow with saliva (no need to administer with liquids).

Administration via feeding tube:

Gastric (eg, NG, G-tube) or post-pyloric tubes (eg, J-tube): Remove plunger from an empty enteral dosing syringe and place orally disintegrating ondansetron tablet(s) into syringe barrel. Replace the syringe plunger and draw up an appropriate volume (eg, 10 to 30 mL) of purified water. Cap the end of the syringe and gently agitate to disperse tablet prior to administration via feeding tube (Ref). **Note:** Some experts suggest that post-pyloric administration is not preferred; consider an alternative dosage form if symptoms do not improve (Ref).

General guidance: Hold enteral nutrition during ondansetron administration (Ref). Flush feeding tube with the lowest volume of purified water necessary to clear the tube prior to administration based on size of patient and/or feeding tube (eg, neonates: 1 to 3 mL; infants and children: 2 to 5 mL; adolescents: 15 mL); refer to institutional policies and procedures (Ref). Following administration, rinse enteral dosing syringe with purified water; draw up rinse and administer contents to ensure delivery of entire dose (Ref). Flush feeding tube with an appropriate volume of purified water and restart enteral nutrition (Ref).

Oral solution (commercially available): Administer with an accurate measuring device (eg, calibrated oral syringe or measuring cup); do not use a household teaspoon or tablespoon to measure dose (overdosage may occur).

Administration via feeding tube:

Gastric (eg, NG, G-tube) or post-pyloric tubes (eg, J-tube): Dilute dose with at least an equivalent volume of purified water prior to administering to reduce osmolality and viscosity; some experts recommend diluting in a volume of purified water that is 3 times the ondansetron solution volume (eg, 10 mL ondansetron solution diluted in up to 30 mL purified water). Draw up diluted solution into enteral dosing syringe and administer via feeding tube (Ref).

Dosage form information: One undiluted formulation has been reported to have an osmolality of 2,935 mOsm/kg (Ref); oral solutions with an osmolality >600 mOsm/kg may increase the probability of adverse GI effects (eg, diarrhea, cramping, abdominal distention, slowed gastric emptying), particularly if administered post-pylorically, inadequately diluted, and/or used in at-risk patients (eg, neonates and infants, patients with short bowel syndrome) (Ref).

General guidance: Hold enteral nutrition during ondansetron administration (Ref). Flush feeding tube with the lowest volume of purified water necessary to clear the tube prior to administration based on size of patient and/or feeding tube (eg, neonates: 1 to 3 mL; infants and children: 2 to 5 mL; adolescents: 15 mL); refer to institutional policies and procedures (Ref). Following administration, rinse container used for preparation with purified water; draw up rinse and administer contents to ensure delivery of entire dose (Ref). Flush feeding tube with an appropriate volume of purified water and restart enteral nutrition (Ref).

Tablets:

Administration via feeding tube:

Gastric (eg, NG, G-tube) or post-pyloric tubes (eg, J-tube): Crush tablet(s) into a fine powder and disperse in at least 10 mL purified water immediately prior to administration; draw up mixture into enteral dosing syringe and administer via feeding tube (Ref).

Dosage form information: Some tablets may be film-coated; administration of film-coated ondansetron tablets via feeding tube may increase the risk of clogging the tube; if used, ensure tablets are sufficiently dispersed prior to administration (Ref).

General guidance: Hold enteral nutrition during ondansetron administration (Ref). Flush feeding tube with the lowest volume of purified water necessary to clear the tube prior to administration based on size of patient and/or feeding tube (eg, neonates: 1 to 3 mL; infants and children: 2 to 5 mL; adolescents: 15 mL); refer to institutional policies and procedures (Ref). Following administration, rinse container used for preparation with purified water; draw up rinse and administer contents to ensure delivery of entire dose (Ref). Flush feeding tube with an appropriate volume of purified water and restart enteral nutrition (Ref).

Parenteral :

IV:

Intermittent infusion:

Chemotherapy-induced nausea and vomiting: Infuse over 15 minutes.

Cyclic vomiting syndrome: Infuse over 15 to 30 minutes (Ref).

IV push: Gastroenteritis, postoperative nausea/vomiting (PONV): May be administered undiluted IV over 2 to 5 minutes (Ref).

IM: Adolescents: PONV: Administer as undiluted injection.

Preparation for Administration: Pediatric

Parenteral: **Note:** Preparation may vary by indication.

IV:

Chemotherapy-induced nausea and vomiting: Dilute prior to infusion. In pediatric patients weighing ≤ 10 kg and/or between 6 months and 1 year of age, may dilute dose in 10 to 50 mL D5W or NS, depending on fluid needs of the patient. In older patients and adults, dilute dose in 50 mL D5W or NS. Concentrations as high as 1 mg/mL have been reported in the literature (Ref).

Cyclic vomiting syndrome: Dilute in 50 to 100 mL of D5W (Ref).

Storage/Stability

Oral disintegrating tablet: Store at 2°C to 30°C (36°F to 86°F).

Oral soluble film: Store at 20°C to 25°C (68°F to 77°F). Store pouches in cartons; keep film in individual pouch until ready to use.

Oral solution: Store upright in carton between 15°C and 30°C (59°F and 86°F). Protect from light.

Tablet: Store at 2°C to 30°C (36°F to 86°F). Protect from light.

Prefilled single use syringe: Store at 20°C to 25°C (68°F to 77°F) or in refrigerator at 2°C to 8°C (36°F to 46°F); protect from light.

Vial: Store at 2°C to 30°C (36°F to 86°F). Protect from light. Chemically and physically stable when mixed in D5W or NS for 48 hours at room temperature; however, diluents generally do not contain a preservative and sterile precautions should be observed. After dilution, do not use beyond 24 hours.

Premixed bag in D5W: Store at 20°C to 25°C (68°F to 77°F), excursions permitted from 15°C to 30°C (59°F to 86°F); may refrigerate; avoid freezing and excessive heat; protect from light.

Use: Labeled Indications

Cancer chemotherapy-induced nausea and vomiting:

IV: Prevention of nausea and vomiting associated with initial and repeat courses of emetogenic cancer chemotherapy (including high-dose cisplatin).

Oral:

Prevention of nausea and vomiting associated with highly emetogenic cancer chemotherapy (including cisplatin ≥ 50 mg/m²).

Prevention of nausea and vomiting associated with initial and repeat courses of moderately emetogenic cancer chemotherapy.

Postoperative nausea and/or vomiting: IV, IM, Oral: Prevention of postoperative nausea and/or vomiting (PONV). If nausea/vomiting occur in a patient who had not received prophylactic ondansetron, IV ondansetron may be administered to prevent further episodes.

Limitations of use: Routine prophylaxis for PONV in patients with minimal expectation of nausea and/or vomiting is not recommended, although use is recommended in patients when nausea and vomiting must be avoided in the postoperative period, even if the incidence of PONV is low.

Radiotherapy-associated nausea and vomiting: Oral: Prevention of nausea and vomiting associated with radiotherapy in patients receiving either total body irradiation, single high-dose fraction to the abdomen, or daily fractions to the abdomen.

Use: Off-Label: Adult

Carcinoid syndrome-associated diarrhea, severe, refractory; Gastroparesis, symptomatic treatment of nausea and vomiting (alternative agent in patients with persistent symptoms refractory to prokinetic therapy); Nausea and/or vomiting, acute, severe; Postoperative nausea and vomiting, treatment or rescue therapy; Pregnancy-associated nausea and vomiting, severe or persistent; Vertigo-associated nausea and vomiting

Medication Safety Issues

Sound-alike/look-alike issues:

Ondansetron may be confused with dolasetron, granisetron, palonosetron

Zofran may be confused with Zantac, Zosyn

Older Adult: High-Risk Medication:

Ondansetron is identified in the Screening Tool of Older Person's Prescriptions (STOPP) criteria as a potentially inappropriate medication in older adults (≥ 65 years of age) with QTc prolongation (O'Mahony 2023).

Metabolism/Transport Effects

Substrate of CYP1A2 (Minor), CYP2C9 (Minor), CYP2D6 (Minor), CYP2E1 (Minor), CYP3A4 (Minor), P-glycoprotein (Minor); **Note:** Assignment of Major/Minor substrate status based on clinically relevant drug interaction potential

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within "CYP3A4 Inducers [Strong]" are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the "Launch drug interactions program" link above.

Amiodarone: May increase QTc-prolonging effects of Ondansetron. Management: Consider alternatives to this combination. If combined, monitor for QTc interval prolongation and ventricular arrhythmias. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk D: Consider Therapy Modification*

Amisulpride (Oral): May increase QTc-prolonging effects of Ondansetron. *Risk C: Monitor*

Apomorphine: Antiemetics (5HT₃ Antagonists) may increase hypotensive effects of Apomorphine. *Risk X: Avoid*

CYP3A4 Inducers (Strong): May decrease serum concentrations of Ondansetron. *Risk C: Monitor*

Dabrafenib: Ondansetron may increase QTc-prolonging effects of Dabrafenib. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Domperidone: May increase QTc-prolonging effects of Ondansetron. *Risk X: Avoid*

Fluorouracil Products: Ondansetron may increase QTc-prolonging effects of Fluorouracil Products. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Haloperidol: Ondansetron may increase QTc-prolonging effects of Haloperidol. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Levoketoconazole: Ondansetron may increase QTc-prolonging effects of Levoketoconazole. *Risk X: Avoid*

MetFORMIN: Ondansetron may increase serum concentrations of MetFORMIN. *Risk C: Monitor*

Pentamidine (Systemic): May increase QTc-prolonging effects of Ondansetron. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Pimozide: May increase QTc-prolonging effects of Ondansetron. *Risk X: Avoid*

Piperaquine: Ondansetron may increase QTc-prolonging effects of Piperaquine. *Risk X: Avoid*

QT-prolonging Agents (Highest Risk): May increase QTc-prolonging effects of Ondansetron. Management: Consider alternatives to this combination. If combined, monitor for QTc interval prolongation and ventricular arrhythmias. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk D: Consider Therapy Modification*

QT-prolonging Antidepressants (Moderate Risk): Ondansetron may increase QTc-prolonging effects of QT-prolonging Antidepressants (Moderate Risk). Ondansetron may increase serotonergic effects of QT-prolonging Antidepressants (Moderate Risk). This could result in serotonin syndrome. Management: Monitor for QTc interval prolongation, ventricular arrhythmias, and serotonin syndrome when these agents are combined. Patients with additional risk factors for QTc prolongation or serotonin syndrome may be at even higher risk. *Risk C: Monitor*

QT-prolonging Antipsychotics (Moderate Risk): Ondansetron may increase QTc-prolonging effects of QT-prolonging Antipsychotics (Moderate Risk). Management: Monitor for QTc interval prolongation, ventricular arrhythmias, including torsades de pointes, when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-prolonging Class IA Antiarrhythmics (Highest Risk): May increase QTc-prolonging effects of Ondansetron. Management: Consider alternatives to this combination. If combined, monitor for QTc interval prolongation and ventricular arrhythmias. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk D: Consider Therapy Modification*

QT-prolonging Class IC Antiarrhythmics (Moderate Risk): May increase QTc-prolonging effects of Ondansetron. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-prolonging Class III Antiarrhythmics (Highest Risk): May increase QTc-prolonging effects of Ondansetron. Management: Consider alternatives to this combination. If combined, monitor for QTc interval prolongation and ventricular arrhythmias. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk D: Consider Therapy Modification*

QT-Prolonging Inhalational Anesthetics (Moderate Risk): Ondansetron may increase QTc-prolonging effects of QT-Prolonging Inhalational Anesthetics (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias, including torsades de pointes when these agents

are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-prolonging Kinase Inhibitors (Moderate Risk): Onāansetron may increase QTc-prolonging effects of QT-prolonging Kinase Inhibitors (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-prolonging Miscellaneous Agents (Moderate Risk): May increase QTc-prolonging effects of Onāansetron. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-prolonging Moderate CYP3A4 Inhibitors (Moderate Risk): Onāansetron may increase QTc-prolonging effects of QT-prolonging Moderate CYP3A4 Inhibitors (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-prolonging Quinolone Antibiotics (Moderate Risk): Onāansetron may increase QTc-prolonging effects of QT-prolonging Quinolone Antibiotics (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-prolonging Strong CYP3A4 Inhibitors (Moderate Risk): May increase QTc-prolonging effects of Onāansetron. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Serotonergic Agents (High Risk): Onāansetron may increase serotonergic effects of Serotonergic Agents (High Risk). This could result in serotonin syndrome. Management: Monitor for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability, mental status changes) when these agents are combined. *Risk C: Monitor*

Sertindole: Onāansetron may increase QTc-prolonging effects of Sertindole. *Risk X: Avoid*

Tapentadol: Onāansetron may decrease analgesic effects of Tapentadol. *Risk C: Monitor*

Thioridazine: Onāansetron may increase QTc-prolonging effects of Thioridazine. *Risk X: Avoid*

TraMADol: Onāansetron may decrease therapeutic effects of TraMADol. Onāansetron may increase serotonergic effects of TraMADol. This could result in serotonin syndrome. Management: Monitor for signs and symptoms of serotonin syndrome/serotonin toxicity (eg, hyperreflexia, clonus, hyperthermia, diaphoresis, tremor, autonomic instability, mental status changes) and diminished tramadol efficacy when these agents are combined. *Risk C: Monitor*

Food Interactions

Tablet: Food slightly increases the extent of absorption. Management: Administer without regard to meals.

Pregnancy Considerations

Onāansetron crosses the placenta (Elkomy 2014; Siu 2006).

Onāansetron can be detected in fetal tissue (Siu 2006). The risk of developing a major congenital malformation following first trimester exposure is under study. Risks related to specific birth defects (eg, cardiac anomalies, oral clefts) require confirmation; available human data are conflicting; however, absolute risk appears to be low (ACOG 2018; Cao 2022; Kaplan 2019; Lemon 2020; Lavecchia 2018; Masarwe 2023; Picot 2020). Clearance is decreased immediately after birth in neonates exposed to onāansetron in utero (Elkomy 2014).

Due to pregnancy-induced physiologic changes, clearance of onāansetron may increase as pregnancy progresses (Lemon 2016). Dose adjustment is not needed when administered for the prevention of

nausea and vomiting associated with cesarean delivery (Elkomy 2014).

Onāansetron may be considered for the treatment of severe or persistent nausea and vomiting of pregnancy (NVP) when preferred agents have failed (ACOG 2018; SOGC [Campbell 2016]). Until additional information related to fetal safety is available, current guidelines suggest use prior to 10 weeks gestation be individualized (ACOG 2018). Dose-dependent QT-interval prolongation can occur with use (Drögemöller 2022); therefore, ECG monitoring is recommended in patients with risk factors for arrhythmia; this may include patients with electrolyte abnormalities associated with some cases of NVP (ACOG 2018).

Onāansetron may be considered as part of a multimodal approach to prevent nausea and vomiting associated with cesarean delivery. A combination of ≥ 2 antiemetics with different mechanisms of action is recommended to treat intraoperative and postoperative nausea and vomiting (Bollag 2021; Griffiths 2012; Habib 2013; Jetling 2017; Macones 2019; Zhou 2018).

An international consensus panel recommends that 5-HT₃ antagonists (including onāansetron) can be used when necessary in pregnant patients receiving chemotherapy for the treatment of gynecologic cancers (Amant 2019).

Breastfeeding Considerations

Onāansetron is present in human milk.

Data are available from a study that developed a physiologically based pharmacokinetic model to predict the presence of drugs in breast milk. Model-predicted data were then compared to samples obtained from lactating patients taking onāansetron:

- The CUDDLE study included 78 women 1 day postpartum treated for postoperative nausea and vomiting with IV onāansetron 4 mg (single dose, n = 55), or 4 to 8 mg in multiple doses (n = 23). A total of 54 breast milk samples and 67 maternal plasma samples were obtained. The median time between the maternal dose and sampling was 24.4 hours (breast milk) and 22.4 hours (plasma). Breast milk and plasma were not collected at standardized times. The pharmacokinetic model predicted the median maximum breast milk concentration of onāansetron to be 20.2 ng/L occurring 0.5 hours after the maternal dose. Predicted data matched those observed from maternal samples. Authors of the study calculated the estimated infant dose of onāansetron via breast milk to be 0.005 mg/kg/day and a relative infant dose (RID) of 3.3% compared to the weight-adjusted maternal dose. Adverse events were not observed in the breastfed infants (Job 2022).
- Per manufacturer's labeling, plasma samples were also obtained from 20 infants in the study exposed to onāansetron via breast milk. Seven infants had onāansetron plasma concentrations below the limit of quantification. In the remaining 13 infants, the reported median onāansetron plasma concentration was 0.78 ng/mL, which is ~ 10 times lower than observed levels in pediatric patients treated with a single therapeutic dose of IV onāansetron.

In general, breastfeeding is considered acceptable when the RID of a medication is $<10\%$ (Anderson 2016; Ito 2000).

Dietary Considerations

Some products may contain phenylalanine.

Monitoring Parameters

ECG if applicable (eg, high-risk or older patients, concurrent use of other medications known to prolong QT interval, electrolyte abnormalities [hypokalemia or hypomagnesemia], heart failure, bradyarrhythmias, and cumulative high-dose anthracycline therapy); serum potassium and magnesium levels. Monitor for signs/symptoms of serotonin syndrome and hypersensitivity; monitor for decreased bowel activity (particularly in patients at risk for bowel obstruction). Monitor for signs/symptoms of myocardial ischemia (during and after administration).

Mechanism of Action

Onāansetron is a selective 5-HT₃-receptor antagonist which blocks serotonin, both peripherally on vagal nerve terminals and centrally in the chemoreceptor trigger zone

Pharmacokinetics (Adult Data Unless Noted)

Onset of action: ~30 minutes

Absorption: Oral: 100%; nonlinear absorption occurs with increasing oral doses; Zofran ODT tablets are bioequivalent to Zofran tablets; absorption does not occur via oral mucosa

Distribution: V_d :

Infants and Children: Surgical patients:

1 to 4 months: 3.5 L/kg

5 to 24 months: 2.3 L/kg

3 to 12 years: 1.65 L/kg

Children and Adolescents: Cancer patients: 4 to 18 years: 1.9 L/kg

Adults: 1.9 L/kg

Protein binding, plasma: 70% to 76%

Metabolism: Extensively hepatic via hydroxylation, followed by glucuronide or sulfate conjugation; CYP1A2, CYP2D6, and CYP3A4 substrate; some demethylation occurs

Bioavailability: Oral: 50% to 70% due to some first-pass metabolism; in cancer patients (adults), 85% to 87% bioavailability possibly related to changes in metabolism

Half-life elimination:

Children: Cancer patients: Children and Adolescents: 4 to 18 years: 2.8 hours; Surgical patients: Infants 1 to 4 months: 6.7 hours; Infants and Children 5 months to 12 years: 2.9 hours

Adults: 3 to 6 hours; Mild-to-moderate hepatic impairment (Child-Pugh classes A and B): 12 hours; Severe hepatic impairment (Child-Pugh class C): 20 hours

Time to peak: Oral: ~2 hours; Oral soluble film: ~1 hour

Excretion: Urine (44% to 60% as metabolites, ~5% as unchanged drug); feces (~25%)

Clearance:

Cancer patients: Children and Adolescents 4 to 18 years: 0.599 L/kg/hour

Surgical patients: Infants and Children: 1 to 4 months: 0.401 L/kg/hour; 5 to 24 months: 0.581 L/kg/hour; 3 to 12 years: 0.439 L/kg/hour

Adult (normal): 19 to 40 years: 0.381 L/kg/hour; 61 to 74 years: 0.319 L/kg/hour; >75 years: 0.262 L/kg/hour

Pharmacokinetics: Additional Considerations (Adult Data Unless Noted)

Altered kidney function: Mean plasma clearance is reduced by 41% (IV) and 50% (oral) in patients with severe renal impairment ($\text{CrCl} < 30 \text{ mL/minute}$).

Hepatic function impairment: In patients with mild to moderate impairment, clearance is reduced 2-fold and the mean half-life is increased to 11.6 hours (compared to 5.7 hours in subjects with normal hepatic function). Clearance is reduced 2- to 3-fold and the apparent V_d is increased, and the half-life is increased to 20 hours in patients with severe hepatic impairment (Child-Pugh class C).

Older adult: In elderly patients >75 years of age, there is a reduction in clearance and an increase in elimination half-life.

Sex: The extent and rate of absorption is greater in women than in men. There is slower clearance, a smaller volume of distribution, and higher bioavailability in women.

Patient Counseling Points

(For additional information see "Ondansetron: Patient drug information")

What is this drug used for?

- It is used to treat or prevent upset stomach and throwing up.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

- Headache
- Feeling tired or weak
- Diarrhea or constipation
- Feeling sleepy
- Anxiety

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

- Trouble passing urine
- Dizziness
- Sudden chest pain or tightness
- Abnormal heartbeat (prolonged QT interval) like a fast or abnormal heartbeat, or if you pass out
- Serotonin syndrome like agitation; change in balance; confusion; hallucinations; fever; fast or abnormal heartbeat; flushing; muscle twitching or stiffness; seizures; shivering or shaking; sweating a lot; severe diarrhea, upset stomach, or throwing up; or very bad headache
- Stomach pain or swelling of the stomach area
- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Aurodanz | Danstro | Imatox | Ondansetron | Tesnad | Vomicure | Vomran | Zofran;

- (AR) Argentina: Cetron | Dantenk | Dasentron | Emivox | Espasevit | Finaber | Finoxi | Nausebron | Ondansetron fabra | Ondansetron filaxis | Ondansetron gobbi | Ondansetron kabi | Ondansetron lazar | Ondansetron lkm | Ondansetron martian | Ondansetron northia | Ondansetron richet | Ondansetron vannier | Ondansetron veinfar | Ondasentron Richet | Setron | Zofran;
- (AT) Austria: Ondansan | Ondansetron | Ondansetron accord | Ondansetron actavis | Ondansetron b braun | Ondansetron b. braun | Ondansetron bluefish | Ondansetron hikma | Ondansetron interpharm | Ondansetron kabi | Ondansetron sandoz | Ondansetron stada | Zofran | Zofran zydis;
- (AU) Australia: Apo ondansetron | Apx ondansetron | Ondansetron | Ondansetron alphapharm | Ondansetron an | Ondansetron apotex | Ondansetron claris | Ondansetron drla | Ondansetron gh | Ondansetron lupin | Ondansetron mylan | Ondansetron mylan odt | Ondansetron odt drla | Ondansetron odt viatris | Ondansetron RI | Ondansetron sz | Ondansetron sz odt | Ondansetron viatris | Ondaz | Onsetron | Zilfojim | Zilfojim ODT | Zofran | Zondan | Zotren odt;
- (BD) Bangladesh: Anset | Apulset | Avona | Emeren | Emistat | Emiston | Nauseb | Ofmit | Ofra | Onaseron | Onaset | Oncodex | Ondagen | Ondamax | Ondason | Onsat | Periset | Puketil | Seton | Setronax | Vomiset;
- (BE) Belgium: Ondansetron | Ondansetron accord | Ondansetron eg | Ondansetron hikma | Ondansetron sandoz | Ondansetron teva | Ondansetron viatris | Zofran | Zofsetron;
- (BF) Burkina Faso: Ondamac | Ondansetron denk;
- (BG) Bulgaria: Emetron | Ondansetron b.braun | Zofran;
- (BR) Brazil: Ansetron | Bienn | Bienn direct | Clor ondansetron | Cloridrato de ondansetrona | Cloridrato de ondansetrona di hidratado | Cloridrato de ondansetrona dihidratado | Emistop | Enavo | Enavo odt | Enjovix flash | Injectrax | Jofix | Listo | Modifical | Nantron | Naudan odt | Nausebron | Nautex | Ondamed | Ondansetrona | Ondavon | Ondif | Ondrau | Ondset odt | Ontrax | Setronax | Volig | Vonau flash | Vonset | Zofran | Zonax;
- (CH) Switzerland: Ondansetron accord | Ondansetron b. braun | Ondansetron Braun | Ondansetron fresenius | Ondansetron labatec | Ondansetron Mepha | Ondansetron nycomed | Ondansetron odt sandoz | Ondansetron sandoz | Ondansetron teva | Ondansetron Teva | Zofran;
- (CI) Côte d'Ivoire: On.setron denk | Onset | Setronax;
- (CL) Chile: Amilene | Antivon | Emetix | Gardoton | Izofran | Levrox | Odatron | Ondansetron | Ondansetrona | OTOC | Trorix;
- (CN) China: An si xin | Ang dan tong | Di li xin | Emeset | En nuo ping | Fu mi ting | Ondansetron | Ondansetron hydrochloride and glucose | Ou bei | Shu dan | Shun ji | Wei ze | Ya bang bei wei | Yi feng | Zofran;
- (CO) Colombia: Aurodanz | Bryterol | Cystron | Emenorm | Modifical | Omiwel | Oncoemet | Ondansetron | Ondax | Ostasyn | Vonau flash | Zofran;
- (CR) Costa Rica: Modifical | On.setron denk;
- (CZ) Czech Republic: Emeset | Emetron | Novetron | Ondansetron Braun | Ondansetron kabi | Ondansetron ratiopharm | Ondansetron sandoz | Ondemet | Zofran;
- (DE) Germany: Axisetron | Cellondan | Ondansetron | Ondansetron 1 a pharma | Ondansetron accord | Ondansetron Aurobindo | Ondansetron Aurus | Ondansetron b.braun | Ondansetron beta | Ondansetron bluefish | Ondansetron carinopharm | Ondansetron DeltaSelect | Ondansetron dura | Ondansetron ever pharma | Ondansetron Hameln | Ondansetron hexal | Ondansetron kabi | Ondansetron Lindopharm | Ondansetron Orca | Ondansetron pfizer | Ondansetron ratiopharm | Ondansetron sandoz | Ondansetron stada | Ondansetron Teva | Ondansetron tillomed | Ondansetron Vipharm | Ondansetron vitane | Ondansetron winthrop | Ondatron | Onsetron denk | Sigondan | Zofran;
- (DO) Dominican Republic: Modifical | Ondansetron | Onone | Setron | Zofran;
- (EC) Ecuador: Antivon | Bryterol | Modifical | Oncoemet | Ondansetron | Ontrona | Setropan | Vonau flash | Zofran;
- (EE) Estonia: Emetron | Ondansetron accord | Ondansetron Bmm | Ondansetron claris | Ondansetron sandoz | Zofran;
- (EG) Egypt: Danofran | Danset | Emerest | Idivomtron | Setronomet | Zofran;
- (ES) Spain: Carvyx | Helmine | Ondansetron aristo | Ondansetron Aurovitas | Ondansetron bluefish | Ondansetron Braun | Ondansetron domac | Ondansetron fresenius kabi | Ondansetron Ips | Ondansetron Normon | Ondansetron Pharmacia | Ondansetron ratio | Ondansetron sandoz |

- Onđansetron serraclinics | Onđansetron stada | Onđansetron teva | Yatrox | Zofran;
- (ET) Ethiopia: Domi Up | Oberem | Onđansetron | Onsetron denk | Vomikind;
- (FI) Finland: Onđansetron accord | Onđansetron accordpharma | Onđansetron bluefish | Onđansetron bmm pharma | Onđansetron fresenius kabi | Onđansetron Hameln | Onđansetron hexal | Onđansetron stada | Onđansetron Synthron | Zofran;
- (FR) France: Onđansetron | Onđansetron Aguetant | Onđansetron arrow | Onđansetron biogaran | Onđansetron Dci | Onđansetron eg | Onđansetron Intas | Onđansetron kabi | Onđansetron Qualimed | Onđansetron ratiopharm | Onđansetron sandoz | Onđansetron Teva | Onđansetron winthrop | Onđansetron Zydus | Zophren;
- (GB) United Kingdom: Onđansetron | Onđansetron accord | Onđansetron pfizer | Onđansetron teva | Ondemet | Setofilm | Zofran;
- (GR) Greece: Biosetron | Cruzafen | Dentron | Fedral | Nofail | Onda | Ondameton | Onđansetron Generics | Onđansetron/b.braun | Onđansetron/Generics | Onđansetron/kabi | Ondaren | Setro-dan | Trondamet | Zodatron | Zofron;
- (HK) Hong Kong: Apo onđansetron | Onđansetron Hameln | Onđansetron kabi | Onđansetron sandoz | Onđansetron teva | Pms Onđansetron | Setronax | Zofran;
- (HR) Croatia: Ondanzetron Kabi | Zofran;
- (HU) Hungary: Antivom | Emetron | Onđansetron ebewe | Onđansetron kabi | Onđansetron pfizer | Onđansetron pharmacenter | Onđansetron pliva | Onđansetron sandoz | Onđansetron-z | Oroset | Vomita | Zentron | Zofran;
- (ID) Indonesia: Cedantron | Ceteron | Dansefion | Dantroxal | Frazon | Fudanton | Insetron | Invomit | Kliran | Lametic | Lexatron | Narfoz | Odanostin | Ondane | Onđansetron | Onđansetron HCL | Ondarin | Ondavell | Onetic | Prezinton | Vomceran | Vometrax | Vometrone | Vomigo | Zantrol | Zofran;
- (IE) Ireland: Emizof | Onđansetron | Onđansetron claris | Onđansetron teva | Ondran | Zofran;
- (IL) Israel: Zofran;
- (IN) India: 4 On | Alset | Anset | Bludan | Capset md | Danotran | Danset | Deltron | Dioset md | Domi Up | Eden | Emcef | Emeban | Emekule | Emeran | Emesafe | Emeset | Emestop MD | Emeton | Emetosim | Emigo | Emitino | Emitron | Emitus | Emnil | Emnil md | Emset | Emsetron | Emtron | Entil | Eterna md | Flatron | Flexi | Glendan | Higo | Ht Blok | Isv | Kod md | Lametic | Levom | Meristron | Myset | Myset md | Mytic | N tron md | Naucid | Nauseidon | Nausehex | Nauseron | Nauset | Nausitroy | Neomit | Noem | Novatron | Nuavomin | O set | O.n.d. | Odanse | Odat | Olron | Oncotor | Ondace | Ondai | Ondamac | Ondamed md | Ondanbic | Ondansetran | Ondanz | Ondar | Ondatab | Ondavom | Ondedom | Ondem | Ondem Md | Ondfast md | Ondisolv | Onditron | Ondoprez | Ondoray | Ondot | Onduro | Ondy md | Ondykav md | Onist md | Onkam | Onmed | Ono | Onset | Onsetrin | Onsett | Onsopil | Onstal | Onswift | Ontic | Ontix md | Ontron md | Oran | Oset | Osetron | Osett | Ozotron | Periset | Periset md | Prospect | Rgnil | Satron | Sav md | Seon md | Set nv | Setiset | Setnorm | Setrona | Setronem | Shutem | Stop Em | Trontis | Udan MD | Vegas | Ventron | Vomalthea | Vomiban | Vomigo | Vomihalt | Vomikind | Vomilife | Vomiof | Vomipen | Vomirek | Vomirest | Vomirid md | Vomirob | Vomisave | Vomiset | Vomiset o | Vomiver | Vomiz | Vorast | Vstop | Zenoset | Zofer | Zondan;
- (IQ) Iraq: Ondanz | Ondisterain | Onitryl | Stop vomit;
- (IT) Italy: Belofran | Onđansetron | Onđansetron b.braun | Onđansetron Braun | Onđansetron Crinos | Onđansetron fresenius | Onđansetron hospira | Ondansetrone aurobindo | Ondansetrone Dr. Reddy's | Ondansetrone Hikma | Ondansetrone Mylan | Ondansetrone Ranbaxy | Ondansetrone Teva | Zofran;
- (JO) Jordan: Nordaset | Onda | Onđansetron vianex | Setron-4 | Zemitron | Zofran;
- (JP) Japan: Onđansetron | Zofran;
- (KE) Kenya: Aurodan | Emadon | Emeset | Emitino | Emitron | Onaseron | Onđansetron denk | Onđansetron Hameln | Ondavell | Ondavian | Ondazed | Ondem Md | Onderon | Onsett | Psm vomitino | Vomikind | Zofra odt | Zofran | Zofran zydis;
- (KR) Korea, Republic of: Hana onđansetron | Ondant | Ondaron | Onfran | Onseran | Onsetron | Zapron | Zofran | Zofran zydis;
- (KW) Kuwait: Onda | Onsit | Zofran;
- (LB) Lebanon: Apo onđansetron | Emeset | Nausetron | Nozentrix | Onđansetron | Onđansetron arrow | Onđansetron biogaran | Onđansetron bluefish | Onđansetron hikma | Onđansetron medis

- | Onzetran od | Setron | Trondamet | Vomiton | Zofran;
- (LT) Lithuania: Emetron | Onāansetron | Onāansetron actiofarma | Onāansetron Aurobindo | Onāansetron Bmm | Onāansetron claris | Onāansetron sandoz | Setronon | Zofran;
- (LU) Luxembourg: Onāansetron b.braun | Zofran;
- (LV) Latvia: Emeset | Emetron | Onāansetron claris | Onāansetron kabi | Zofran;
- (MA) Morocco: Chemoset | Onset | Zofran | Zophren;
- (MX) Mexico: Antivon | Dosartron | Emistop | Ht bloc | Krindor | Nevolnost | Onāansetron | Precirux | Setronax | Vylkor | Zincolset | Zofran;
- (MY) Malaysia: Apo onāansetron | Emeset | Zofran;
- (NG) Nigeria: Emitino | Onāansetron | Zofran;
- (NL) Netherlands: Onāansetron | Onāansetron A | Onāansetron actavis | Onāansetron Alphaarma | Onāansetron aspire | Onāansetron bluefish | Onāansetron cf | Onāansetron fresenius kabi | Onāansetron ratiopharm | Onāansetron sandoz | Zofran | Zofran 4 zydis;
- (NO) Norway: Onāansetron | Onāansetron Aurobindo | Onāansetron aurora medical | Onāansetron bluefish | Onāansetron bmm pharma | Onāansetron fresenius kabi | Onāansetron Hameln | Onāansetron mayne | Onāansetron orifarm | Onāansetron pfizer | Zofran;
- (NZ) New Zealand: Apo onāansetron | Onāansetron | Onāansetron claris | Onāansetron dr Reddy | Onāansetron drla | Onāansetron kabi | Onrex | Periset | Periset odt | Zofran;
- (PA) Panama: Modifical | On.setron denk | Onāansetron;
- (PE) Peru: Alencar | Modifical | Onāansetron | Ondapharm | Onmek | Ontrona | Zofran;
- (PH) Philippines: Amnoset | Emistop | Enset 8 | Kabidan | Ondatrix | Onsett | Onsia | Onstal | Onstal forte | Onzet | Vomiton | Zofran;
- (PK) Pakistan: Adosetron | Anomed | Chemset | Danomed | Danset | Dansetron | Deston | Endtron | Nixvom | Odanex | Ondan | Ondanles | Onāansetron | Ondanz | Ondasave | Ondaset | Ondenles | Ondonix | Ongene | Onitron | Onrem | Onseron | Onset | Onvin | Osetron | Preset | Prevon | Sandem | Vemtix | Vominec | Zofran;
- (PL) Poland: Atossa | Ebesetron | Emetron | Ondalek | Onāansetron accord | Onāansetron bluefish | Onāansetron Braun | Onāansetron claris | Onāansetron kabi | Ondaren | Setronon | Zofran;
- (PR) Puerto Rico: Onāansetron | Onāansetron HCL | Onāansetron odt | Zofran;
- (PT) Portugal: Emytron | Nausiend | Ondansetrom | Ondansetrom actavis | Ondansetrom Braun | Ondansetrom Ciclum | Ondansetrom generis | Ondansetrom hikma | Ondansetrom inibsa | Ondansetrom labesfal | Ondansetrom normon | Ondansetrom toLife | Otobrol | Zofran;
- (PY) Paraguay: Espasevit | Kelme flash | Limine | Metasone | Onāansetron fusa | Onāansetron veinfar | Ondatron;
- (QA) Qatar: Dansetron | Nomit | Ondan | Onāansetron Kabi | Ondavell | Onsite | Osetron | Setron | Vommet-Off | Vomiran | Zofran | Zofran Melt;
- (RO) Romania: Osetron | Zofran;
- (RU) Russian Federation: Emeset | Emetron | Latran | Lazaran vm | Onāansetron | Onāansetron teva | Ondasol | Ondator | Ondavell | Osetron | Rondaset | Setronon | Vero onāansetron | Zofran;
- (SA) Saudi Arabia: Apo onāansetron | Emedan | Imatox | Kromafina | Ondanion | Ondans | Onāansetron | Onāansetron kabi | Onāansetron medis | Pms Onāansetron | Zemitron | Zofran | Zofran melt | Zoron;
- (SE) Sweden: Onāansetron 2care4 | Onāansetron abacus medicine | Onāansetron accord | Onāansetron accordpharma | Onāansetron alternova | Onāansetron amneal | Onāansetron Aurobindo | Onāansetron b braun | Onāansetron bluefish | Onāansetron bmm pharma | Onāansetron copyfarm | Onāansetron ebb | Onāansetron eql pharma | Onāansetron fresenius kabi | Onāansetron Hameln | Onāansetron hexal | Onāansetron hospira | Onāansetron mylan | Onāansetron nycomed | Onāansetron orifarm | Onāansetron pfizer | Onāansetron sandoz | Onāansetron stada | Onāansetron teva | Onāansetron vian | Zofran;
- (SG) Singapore: Onāansetron | Onāansetron sandoz | Ondavell | Zofran;
- (SI) Slovenia: Onilat | Setronon | Zofran;
- (SK) Slovakia: Onāansetron b. braun | Onāansetron kabi | Onāansetron sandoz | Onāansetron teva | Ondemet | Onsetrogen | Setron | Zofran;
- (SR) Suriname: Onāansetron | Onāansetron Aurobindo | Onāansetron cf | Onāansetron denk | Onāansetron mylan | Zofran;
- (SV) El Salvador: Antivon | Ondasetron;

- (TH) Thailand: Emistop | Nautah | Onsia | Vomitron | Zetron | Zofran;
- (TN) Tunisia: Ondansetron | Ondansetron Renaudin | Zemitron;
- (TR) Turkey: Nauzex | Ondaren | Onzyd | Santis | Zofer | Zofran | Zoltem | Zontron | Zophralen;
- (TW) Taiwan: Ondan | Sopran | Vomiz | Zofran;
- (UA) Ukraine: Emeset | Emesetron zdorovje | Emetron | Ondaar | Ondansetron | Ondansetron sandoz | Osetron | Setronon | Vomikind md | Zofetron | Zofran;
- (UG) Uganda: Aurodanz | Emadon | Emitino | On.setron denk | Ondanset | Ondansetron denk | Onsett | Vomikind md | Vomistat | Vomiz;
- (UY) Uruguay: Dasentron | Emivox | Izofran | Ondansetron | Ondaset | Ondatie | Quimiofran | Setron;
- (VE) Venezuela, Bolivarian Republic of: Emistop | Ondansetron | Ondasetron | Osetron | Tructum | Vomigon | Zofran;
- (VN) Viet Nam: Dloe | Ondanov | Slandom | Yuhanonseran;
- (ZA) South Africa: Aspetron | Austell ondansetron | Cipla ondansetron | Nauseatron | Ondansetron | Ondansetron cipla | Vomiz | Zofer | Zofran | Zothena;
- (ZM) Zambia: Onsett;
- (ZW) Zimbabwe: Ondansetron fresenius | Onsett | Zofran

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